

Ahtesham Alvi

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EDUCATION

University of Maryland, College Park | College Park, MD
Bachelor of Science, Computer Science (Honors) and Finance
Minors: Data Science, Robotics and Autonomous Systems

Expected May 2027

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C/C++, Rust, OCaml, SQL, SOQL, Apex, Bash
ML & Data: PyTorch, scikit-learn, XGBoost, pandas, NumPy, NetworkX, DuckDB, Tableau, AlphaFold2, RFdiffusion, ProteinMPNN
Web & Systems: React, FastAPI, Flask, REST APIs, WebSockets, Docker, Linux, Git/GitHub, Jenkins, SLURM/HPC, Vercel, Render, ROS2 (Humble), Gazebo, Qiskit
Platforms & AI Tooling: Salesforce, Jira, Confluence, Claude Agent SDK, MCP, OpenAI API, Excel, LaTeX

EXPERIENCE

fusionSpan | *Software Developer Intern* | Rockville, MD

June 2026 – August 2026

- Developed, tested, and deployed production Salesforce CRM solutions for nonprofit and association clients (MHI, MCAA).
- Closed 50+ tickets of client-facing features and bug fixes, tracking work across the team in Jira and Confluence.
- Established an internal code-testing process adopted by the team, standardizing QA before client releases.

Urban Food Alliance | *Software Development Intern* | Remote

July 2025 – November 2025

- Built a financial-literacy chatbot in Python with an NLP intent/response pipeline, increasing user engagement by 20%.
- Improved query understanding and response accuracy through NLP tooling and iterative evaluation of dialogue quality.
- Shipped in a Scrum cadence with Git/GitHub code review and pre-release testing, reducing post-release defects.

RESEARCH

De Novo Protein Binder Design for IBD Therapeutics (IL-23)

March 2026 – Present

- Built a 9-stage GPU pipeline (RFdiffusion, ProteinMPNN, AlphaFold2) on SLURM HPC to design de novo miniprotein binders.
- Screened 50 designs against 4 targets (p19, p19-p40, p40, IL-12) over ~1,000 AlphaFold2 initial-guess predictions, scored with ipSAE, Rosetta binding energy, and PyMOL steric superposition onto the 5MZV receptor complex (median C-alpha RMSD 1.96 Å).
- Sole-authored a paper showing 45/50 designs sterically clash with the obligate p40 subunit despite high predicted interface confidence, establishing assembly compatibility as a confidence-independent validation step (Boltz-2 Spearman rho = -0.60).

Cross-Era Social Bot Detection (CMSC396H) | [GitHub](#)

April 2026 – May 2026

- Engineered account, behavioral, graph (PageRank, reciprocity, ego density), and node2vec features over 4 Twitter/X bot datasets spanning 2015–2022 (up to 1M users, 88M tweets), normalized into one schema in DuckDB, NetworkX, and pandas.
- Trained 6 classifier families across all 16 train/test dataset pairs (96 models) under a per-pair feature contract.
- Co-authored a paper showing within-era AUC of 0.86–1.00 collapses to near chance (0.48–0.66) across eras as bot behavior evolved.

FIRE Bioinspired Robotics – Flapping Winged Glider | [GitHub](#)

May 2024 – July 2024

- Led a flapping-wing glider build: Arduino Nano firmware for torque control and SD-card telemetry; carbon-fiber airframe.

PROJECTS

QuantumGuard – Quantum Key Distribution Demo | [GitHub](#) | [Live Demo](#)

April 2026

- Deployed a full-stack app (React/TypeScript on Vercel, FastAPI + WebSockets on Render) staging a three-party MITM attack.
- Implemented BB84 in Qiskit Aer over 256 qubits, deriving AES-128-GCM keys and rejecting exchanges above 11% QBER.

Exoplanet Habitability Classification ML Model | [GitHub](#)

January 2026 – Present

- Developing neural and embedding-based models over NASA and JPL catalogs to surface habitable exoplanet candidates.

Mini C Compiler (Rust, OCaml) | [GitHub](#)

May 2025

- Built a compiler front end (lexer, parser, AST, constant folding, type inference) for a C-like language in both Rust and OCaml.

Robotic Arm Manipulator Control | [GitHub](#)

August 2025 – December 2025

- Developed an inverse kinematic planner with ArUco vision perception for a commercial robotic arm on ROS2, Docker, and Gazebo.
- Built homography-based pixel-to-world calibration and motion sequencing for autonomous block sorting and stacking.

AST-Based Unix Shell (C) | [GitHub](#)

December 2024

- Implemented a Unix shell in C with an AST execution engine: pipes, &&/||, sequencing, subshells, and I/O redirection.

LEADERSHIP & ADDITIONAL EXPERIENCE

UMD Finance Committee | *Budget Reviewer* | College Park, MD

September 2024 – Present

- Administered a \$2.7M annual budget across 200+ campus organizations; adjudicated appeals and voted on funding legislation.

UMD College of Computer, Mathematical & Natural Sciences | *Office Assistant*

January 2024 – Present

- Archived 4,000+ documents; audited payroll and tax data (W-4) in Workday and Excel to IRS compliance standards.

AI Construction LLC | *Marketing Manager; General Assistant* | Fairless Hills, PA

June 2017 – Present

- Designed the company website and vehicle marketing in Photoshop; remodeled residential kitchens and baths on time and budget.